GreenCart: IDH 3350 Semester Reflection

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1 Project Overview

This semester project focuses on building GreenCart, a practical sustainability platform for students and local community members in Tampa Bay. The goal is to reduce the gap between caring about the environment and making sustainable everyday choices. The live platform is available at green-space-three.vercel.app.

Quick Product Preview

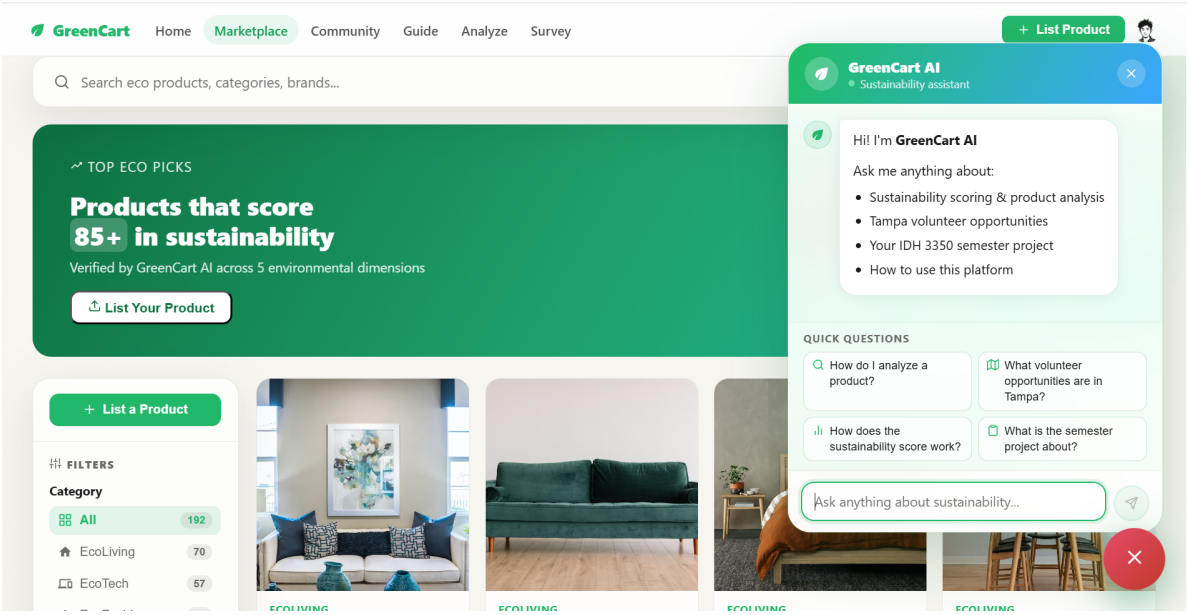


Figure 1: Marketplace view with curated products, filters, and sustainability focus.

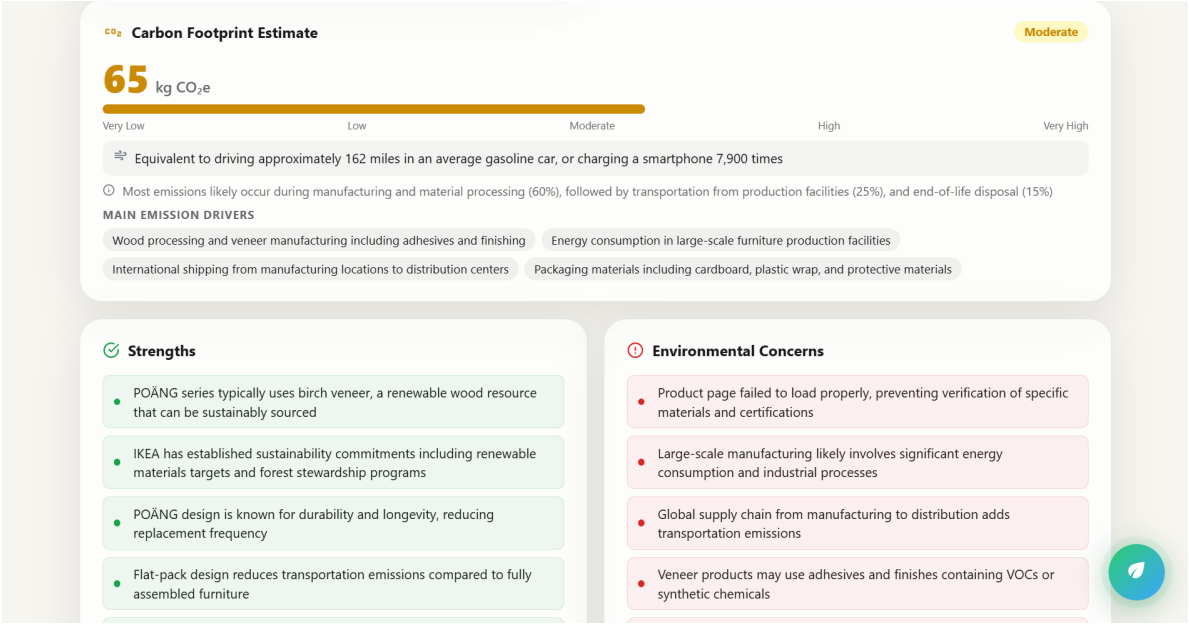


Figure 2: Product analysis view with AI-generated sustainability insights.

2 Documentation of Activity

Over the semester, I built and improved five main features:

- Marketplace with 190+ eco products and filters.
- AI analysis tool that scores product sustainability from a URL.
- Community page for local sharing and volunteer matching.
- Student survey on consumer behavior and sustainability.
- Guide page for onboarding and class project support.

Main tools used: Next.js, React, Supabase, Clerk authentication, and the Anthropic API.

3 Resulting Product

The final product is a working web app that helps users:

1. discover better products,
2. compare sustainability impact,
3. reflect on personal behavior,

4. connect with local environmental opportunities.

This project combines technical implementation and environmental engagement. It was designed around real barriers such as lack of trusted information, convenience, and habit.

4 Reflection

This project was challenging and meaningful. The most important lesson for me was that sustainability is not only a science issue; it is also a design and information problem. If people can get clear and useful information quickly, they are more likely to make better choices.

Building GreenCartchanged how I shop and how I think about impact. I now pay more attention to product lifecycle and alternatives instead of only price or convenience.

I plan to continue improving this project by expanding local community features and improving survey analytics.

5 Class Concept Connection

This work directly connects to the attitude–behavior gap discussed in class. Many people care about sustainability but do not act consistently. GreenCartaddresses this by making sustainable action easier, clearer, and more social.

6 Screenshots and Evidence

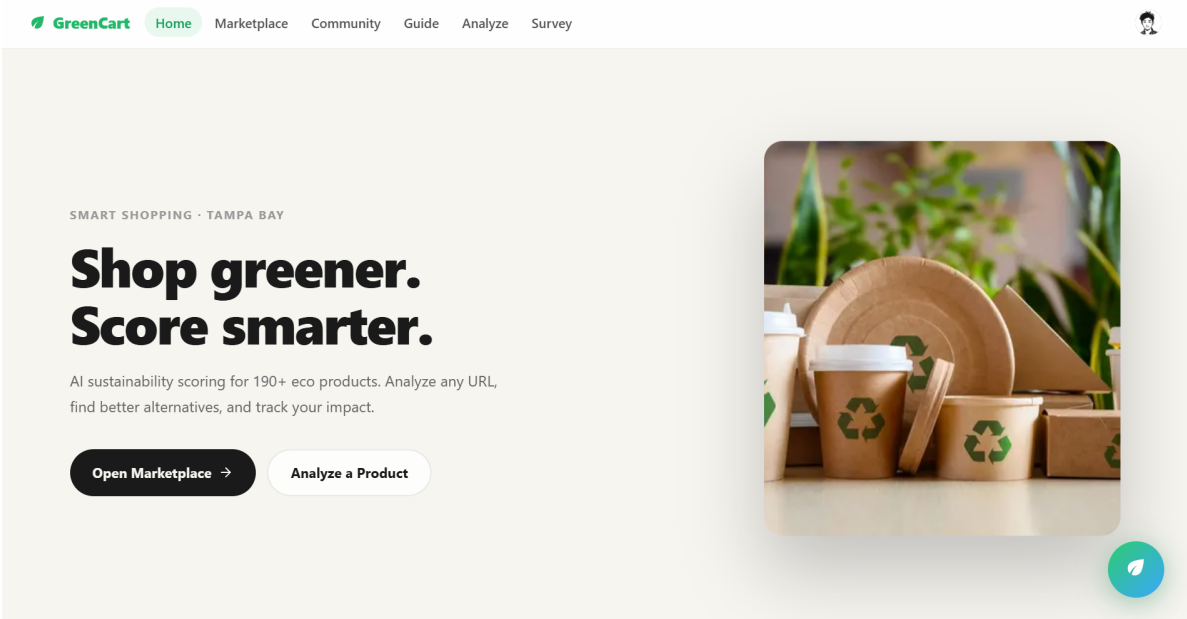


Figure 3: Landing page of GreenCart.

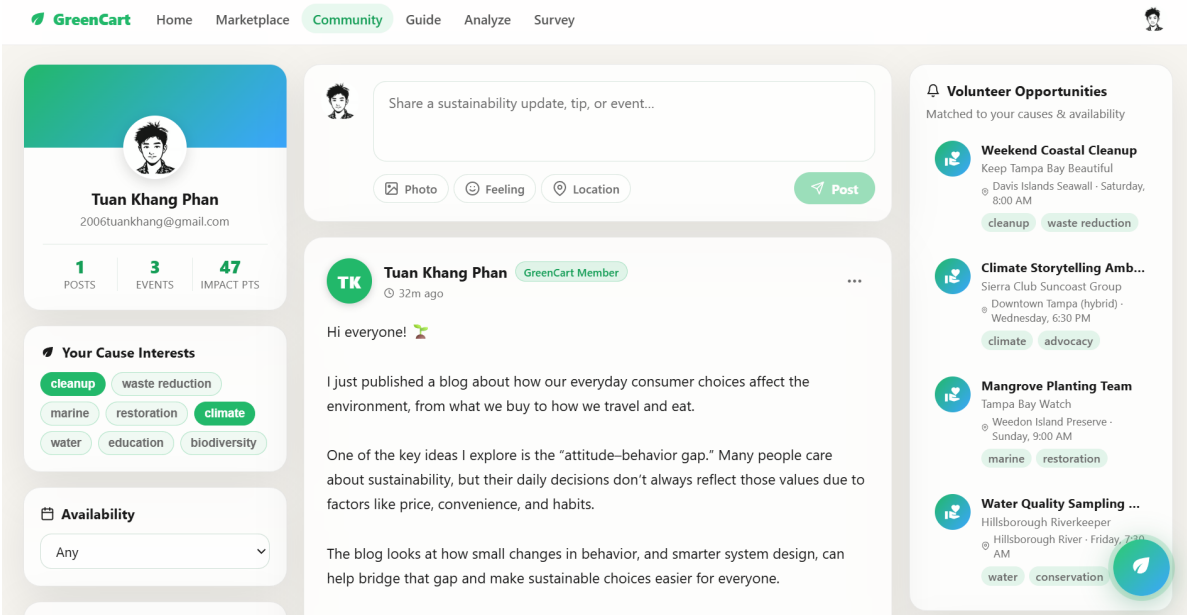


Figure 4: Community page with posting and volunteer matching.

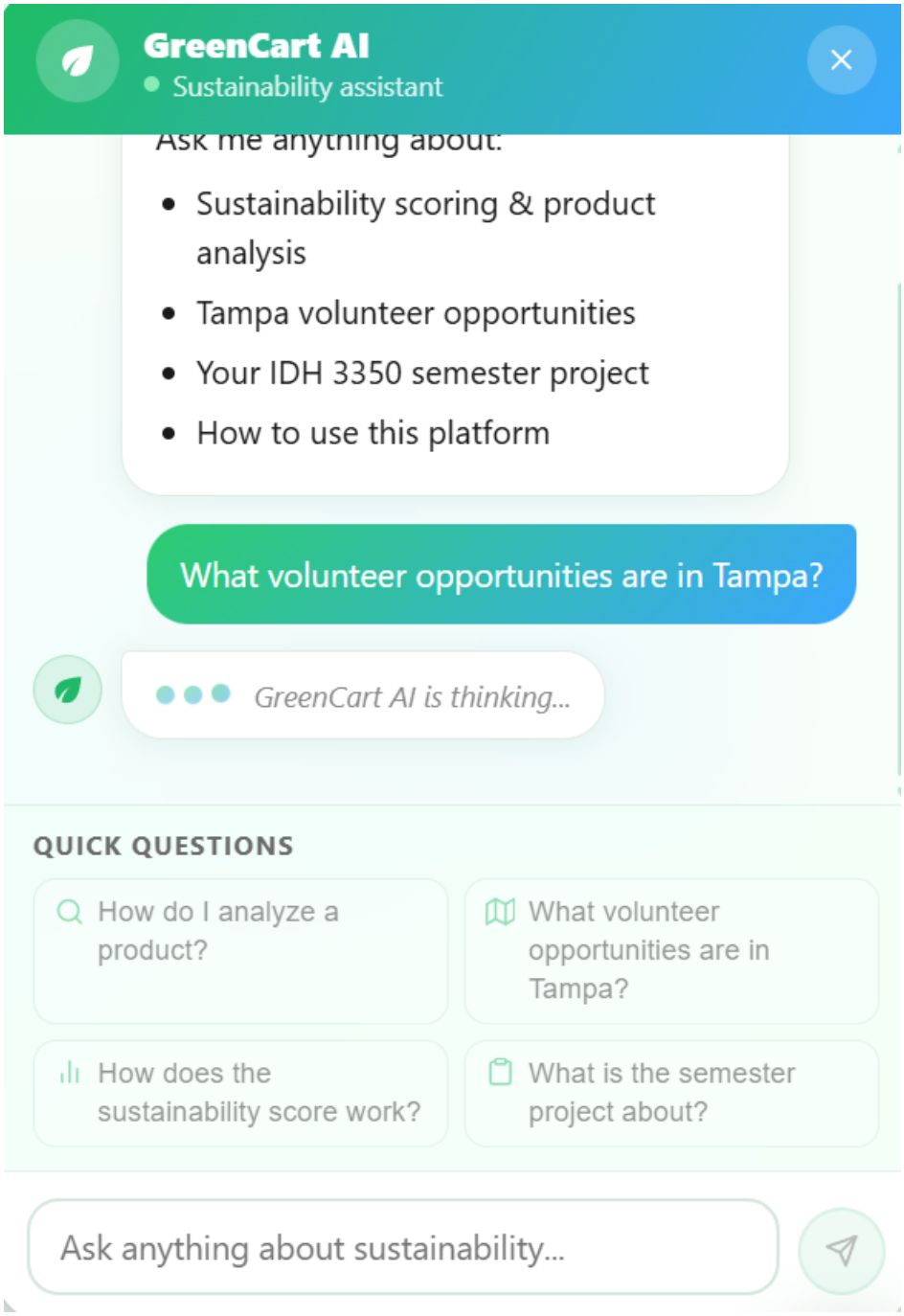


Figure 5: Built-in chatbot assistant for sustainability questions.

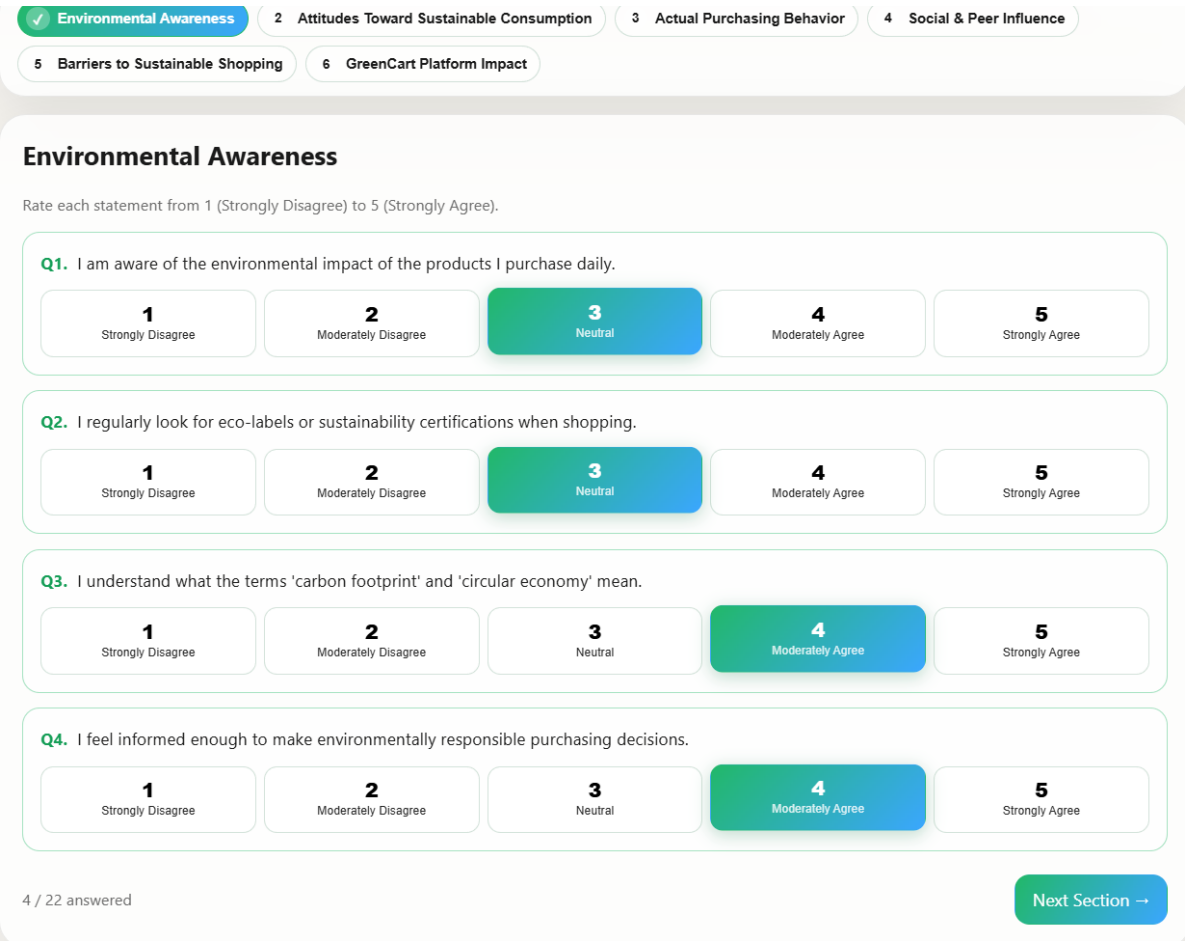
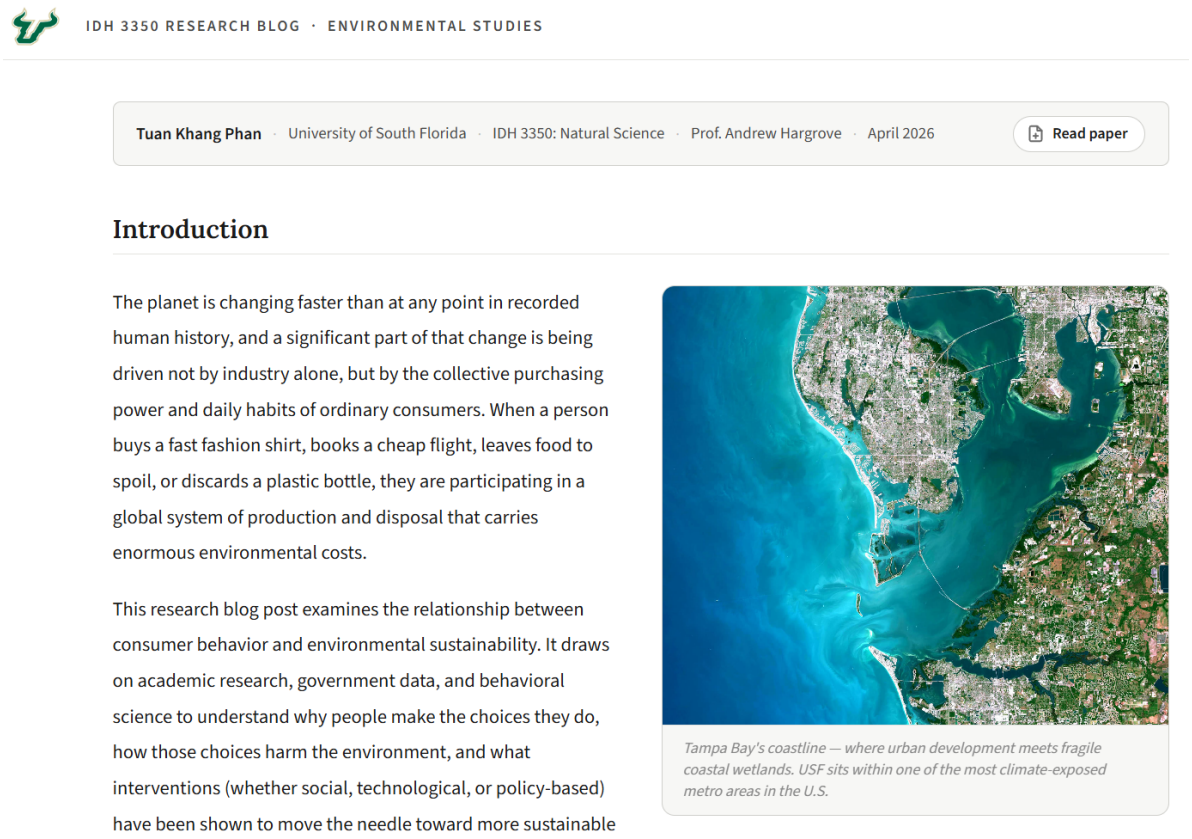


Figure 6: Survey interface on environmental awareness and behavior.



IDH 3350 RESEARCH BLOG · ENVIRONMENTAL STUDIES

Tuan Khang Phan · University of South Florida · IDH 3350: Natural Science · Prof. Andrew Hargrove · April 2026 Read paper

Introduction

The planet is changing faster than at any point in recorded human history, and a significant part of that change is being driven not by industry alone, but by the collective purchasing power and daily habits of ordinary consumers. When a person buys a fast fashion shirt, books a cheap flight, leaves food to spoil, or discards a plastic bottle, they are participating in a global system of production and disposal that carries enormous environmental costs.

This research blog post examines the relationship between consumer behavior and environmental sustainability. It draws on academic research, government data, and behavioral science to understand why people make the choices they do, how those choices harm the environment, and what interventions (whether social, technological, or policy-based) have been shown to move the needle toward more sustainable

Tampa Bay's coastline — where urban development meets fragile coastal wetlands. USF sits within one of the most climate-exposed metro areas in the U.S.

Figure 7: Companion research blog used in the project.

7 Links

- Platform: green-space-three.vercel.app
- Research blog: env-blog.vercel.app